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Ensemble deep learning models for respiratory disease detection using cough analysis

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ABSTRACT

Respiratory monitoring using analysis of cough sound has been identified as a potential, non-intrusive technique for early diagnosis of irregular respiratory patterns. In this work, we introduce an automated system that classifies audio recordings into three groups: unhealthy coughs, healthy coughs, and non-cough sounds. The proposed system begins by extracting advanced acoustic features, particularly Mel Filter Bank (MFB) coefficients combined with a novel architecture integrating preprocessing, feature extraction, and a stacked ensemble of deep learning models. The system achieves impressive results: XGBoost reaches an accuracy of 99.49 %, while our proposed deep fusion model combining CNN, MLP, and RNN architectures pushes accuracy up to 99.94 %. These findings show how effective and reliable the solution is, making it a practical tool for large-scale respiratory health screening with minimal equipment.

Introduction

Respiratory disorders remain a major public health issue, causing around 20 % of global deaths, according to the World Health Organization [17]. Both chronic conditions like COPD, which is expected to become the third leading cause of death by 2030, and acute infections such as pneumonia, responsible for 15 % of child mortality, contribute to this ongoing threat. The recent history of COVID-19, with more than 767 million reported cases globally, has highlighted the urgent need for improved respiratory disease diagnostic platforms.

Early diagnosis remains critical, as clinical evidence demonstrates that screening within 5–7 days of symptom onset can reduce mortality by 60 % in severe pneumonia and by 45 % in COPD exacerbations. However, conventional diagnostic methods, face several limitations, especially in resource-limited settings. Although PCR testing offers high accuracy, it is expensive and turnaround time (24–72 h). Furthermore, advanced imaging technologies are not available in 85 % of hospitals in sub-Saharan Africa.

To overcome these limitations, AI-assisted cough analysis has

emerged as a promising alternative. This method is based on the principle that different respiratory pathologies produce distinct acoustic signatures in cough sounds. Because cough patterns carry valuable information about the respiratory system, AI can help detect subtle features associated with diseases such as COVID-19 [12,14]. The key advantage of this approach is its simplicity and scalability cough sounds can be recorded and analyzed in real time, making it suitable for mass screening. Prior research has explored AI in COVID-19 diagnostics, particularly through imaging modalities such as CT scans and X-rays [3, 41,47,62]. Additionally, several studies have investigated machine learning for analyzing cough, breathing, and speech audio to develop rapid and cost-effective testing methods [9,64]. However, challenges remain in optimizing acoustic feature extraction, improving classification accuracy with ML and DL algorithms, and designing scalable solutions that can be deployed in real-world clinical environments.

The development of our AI-based cough analysis framework is driven by three key unmet needs in respiratory disease diagnostics, speed vs. Accuracy Trade-off, while PCR remains the gold standard, its turnaround time (24–72 h) is impractical for rapid mass screening during

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outbreaks. Resource Inequality, advanced imaging is inaccessible in more than 85 % of sub-Saharan hospitals, creating diagnostic disparities. Limitations of Prior AI Solutions, Existing cough analysis models often use acoustic features or standalone ML/DL models, missing opportunities for performance gains through hybrid feature engineering and ensemble learning. Our work directly tackles these gaps by proposing a deployable, hardware-agnostic solution that combines computational efficiency with clinical-grade accuracy, particularly crucial for pandemic preparedness and equitable healthcare delivery.

To address these challenges, we introduce a novel AI-based framework for automated cough screening. Our system uses a multi-stage pipeline that combines advanced acoustic feature extraction with a stacked ensemble of deep learning and traditional machine learning models. While the approach can be extended to a variety of respiratory diseases, we demonstrate its effectiveness in a targeted case study on COVID-19 diagnosis using the COUGHVID dataset [9], where abnormal coughs are indicative of SARS-CoV-2 infection.

Our primary contributions are as follows:

- **State-of-the-Art Comparative Analysis:** We present a comprehensive evaluation of multiple ML and DL techniques, including CNN, MLP, RNN, RF, and XGBoost, for the detection of COVID-19 coughs.
- **Ensemble Learning for Enhanced Accuracy:** By integrating ensemble learning approaches, we achieve improved model performance compared to individual classifiers.
- **Innovative Audio Feature Modeling:** We demonstrate the effectiveness of using MFB features in COVID-19 cough detection, which outperformed traditional MFCC features.
- **Superior Performance:** Our deep fusion ensemble model combining CNN, MLP, and RNN architectures achieves a remarkable accuracy of 99.94 %, demonstrating the feasibility of reliable, non-invasive COVID-19 detection.

These results emphasize the potential of AI-based cough analysis as a rapid and scalable diagnostic tool, especially in situations where conventional testing methods may be impractical or unavailable.

The remainder of this paper is organized as follows:

[Section 2](#) presents the related works.

[Section 3](#) details the proposed approach, including data pre-processing and model architecture.

[Section 4](#) reports the experimental results and model performance.

[Section 5](#) compares our approach with existing studies.

[Section 6](#) discusses the key findings and limitations.

[Section 7](#) concludes the paper and outlines future directions.

[Section 8](#) lists the references used throughout the paper.

Related works

Since the COVID-19 pandemic began, artificial intelligence (AI) has proven to be a powerful tool for diagnosing infections. Machine learning (ML) and deep learning (DL) techniques have been widely applied to detect COVID-19 using both radiographic images and respiratory sounds. While early studies focused mainly on chest X-rays and CT scans, cough sound analysis has gained attention as a cost-effective and non-invasive option [8,44].

Research into cough-based diagnostics has explored various ML and DL models to classify COVID-19-related coughs. [10] pioneered the approach by using Mel-frequency cepstral coefficients (MFCCs) as input features to distinguish COVID-19 coughs, achieving 80 % accuracy with a dataset of over 20,000 samples. [19] extended this by combining MFCC features with supervised ML models, though they only achieved 57 % accuracy, highlighting challenges with feature-based classification.

To address these issues, deep learning methods such as convolutional neural networks (CNNs) have been explored. [28] achieved 90 % accuracy with a CNN-based model trained on cough spectrograms, using

data augmentation to compensate for the lack of high-quality recordings. Similarly, [2] combined CNNs and recurrent neural networks (RNNs) to capture both spectral and temporal cough features, reaching 85 % accuracy. These studies demonstrate the importance of multimodal feature representation for robust classification. Beyond COVID-19, recent studies have demonstrated that respiratory sounds can be effectively utilized for the automated diagnosis of a broad range of pulmonary diseases using advanced deep learning architectures. The Pulmo-TS2ONN model [53], for instance, employs a triple-scale self-operational neural network combined with time-frequency features to classify seven distinct respiratory conditions, achieving an accuracy of 98.88. Similarly, the VGAResNet [54] framework leverages visibility graph-based representations of lung sound in conjunction with a residual neural network to detect chronic obstructive pulmonary disease (COPD), with reported accuracy, sensitivity, and specificity of 95.13 %, 96.33 %, and 94.37 %, respectively. In the more complex domain of interstitial lung diseases (ILDs), the ILDNet model [52] utilizes a sinc-convolutional residual architecture to classify pathological respiratory sounds, achieving 81.25 % accuracy. In the context of COVID-19, recent work has shown that cough-based classification remains highly effective when combined with advanced feature extraction. For example, [61] proposed the Cough2COVID-19 (MLEDL) framework (2024), which introduces an innovative hybrid approach integrating diverse feature representations, MFCC-MLP, Spectrogram-CNN, and Chromagram-MLP, through the novel CoughFeatureRanker methodology, achieving AUC scores of 0.91, 0.88, and 0.83 on the Cambridge, Virufy, and Coswara datasets, respectively. Similarly, [65] introduced a VGGish-BiLSTM-attention architecture trained on the COUGHVID dataset, reaching an accuracy of 92.41 % and an F1-score of 92.23 %, demonstrating strong generalization capacity despite variability in crowd-sourced data. Collectively, these studies highlight the expanding role of AI in sound-based diagnostics, extending its applicability well beyond COVID-19 and into broader domains of respiratory medicine.

Despite their success, CNNs and RNNs face generalization challenges across diverse populations. As a result, ensemble learning techniques have been increasingly adopted for greater robustness and accuracy. [45] proposed an ensemble model combining CNNs, RNNs, and random forests, achieving 87 % accuracy and 83 % sensitivity on over 30,000 cough recordings by leveraging complementary model strengths to reduce overfitting. [57] further improved ensemble techniques by integrating XGBoost for feature selection and addressing class imbalances, achieving 90 % accuracy with enhanced sensitivity and specificity. This highlights the clinical relevance of ensemble learning, particularly in real-world scenarios with dataset variability and imbalances.

Public datasets such as COUGHVID [43], Coswara [57], and Flusense [11] have been crucial in advancing COVID-19 cough classification research by providing diverse cough samples for model development. However, challenges remain, such as standardizing data, accounting for recording variations, and ensuring clinical validation.

Proposed approach

The proposed approach for classifying COVID-19-related positive and negative cough sounds is structured into two main parts. The first part introduces the datasets utilized in this study, while the second part details the methodologies applied, which leverage machine learning, deep learning, and ensemble learning techniques, as illustrated in [Fig. 1](#).

Dataset selection

Dataset description

The analysis relies on the open-access COUGHVID dataset [43], a crowdsourced collection of cough audio recordings designed to support research on respiratory disease detection. The dataset includes over 25,000 recordings from five continents, with 1155 reporting

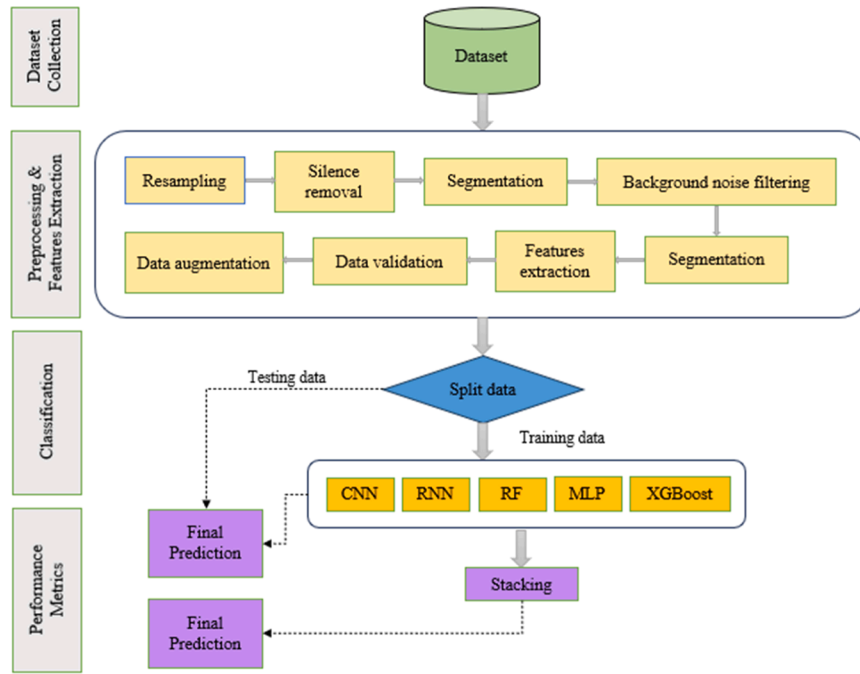


Fig. 1. Architecture of the proposed COVID-19 audio classification system.

COVID-19-related symptoms. In addition to the audio files, clinical data and metadata such as age, sex, and respiratory condition details are provided.

Data collection occurred between April 2020 and October 2021 through a web app hosted at the Swiss Federal Institute of Technology in Lausanne. Users recorded up to 10 s of cough sounds by clicking a “Recordings” button. Participants could optionally complete a metadata form, but all recordings were stored on the server in WEBM or OGG format, regardless of metadata submission. The dataset’s demographic diversity enhances its global representativeness, improving the generalizability of research findings.

Quality control ensured the reliability of audio and metadata. A cough detection model was developed to exclude non-cough recordings, assigning a “cough_detected” probability label to each entry. To further enhance data quality, four experts annotated approximately 1000 recordings each using an online platform, focusing on variables such as quality, cough type, dyspnea, wheezing, stridor, and diagnosis severity. These features, along with questionnaire variables, are detailed in Table 1.

Table 1

Metadata variables.

	Description	Mandatory
datetime	Time and date of the received recording	Yes
cough_detected	Cough sound recording likelihood as determined by an artificial detection algorithm	Yes
SNR	An estimation of the signal-to-noise ratio of the cough recording	Yes
latitude	Self-reported longitude geolocation	No
age	Self-reported age value	
gender	Self-reported gender	
respiratory_condition	The patient has other respiratory conditions	
fever_muscle_pain	The patient has fever or muscle pain	
Status	The patient reports whether they are healthy (healthy), have symptoms but no diagnosis (symptomatic), or have been diagnosed with COVID-19 (COVID).	No

Selection and composition of dataset

Our final dataset combines samples from two sources: the COUGHVID dataset and additional samples from Kaggle. To ensure data homogeneity, we standardized all audio files to a common audio format and applied a uniform preprocessing pipeline, including amplitude normalization and noise filtering.

Following the selection criteria from the questionnaire responses, we selected recordings from the COUGHVID dataset based on two strict conditions: confirmed COVID-19 status (`covid_status = 1`) and detection of a positive cough (`cough_detected = 1`). This resulted in 684 recordings of confirmed COVID-19 coughs.

In contrast, healthy cough and non-cough samples were obtained exclusively from the Kaggle dataset. In order to guarantee the dataset’s dependability, we restricted healthy samples to recordings collected before 2019, ensuring reliable negative labels. This method increases the legitimacy of our labeling procedure.

Our database is distributed as follows:

- 684 confirmed COVID-19 cough recordings from COUGHVID.
- 12,479 healthy cough samples from Kaggle.
- 697 non-cough samples from Kaggle.

The dataset’s reported demographic distribution is exclusive to the COUGHVID recordings. The participants in these recordings, who are from over 121 countries, are 36.5 years old on average, 66 % male, and 33.5 % female. In the parts that follow, the process for managing dataset imbalance will be thoroughly described.

Dataset selection

The most important step in managing audio recordings is data preparation, which prepares raw data for effective and efficient analysis. Enhancing the quality of the data and extracting pertinent features for machine learning algorithms are the goals of several steps in this procedure. Raw audio capture, resampling, silence removal, noise reduction, segmentation, background noise filtering, normalization, feature extraction, audio file organization, data validation, and data augmentation as outlined in this section.

RAW-Audio

The initial step involves collecting raw data in the form of audio recordings, typically captured using microphones or other sound-capturing devices. These recordings encompass a broad spectrum of information, including target sounds, background noise, and silent intervals. Given the unstructured and noisy nature of raw data, pre-processing is necessary to transform these audio files into a structured format suitable for direct application to machine learning models.

Resampling

The signal processing stage, following the collection of audio data, begins with resampling at 16 kHz. Standardizing the sampling rate ensures audio consistency and enhances processing efficiency. This frequency preserves essential sound characteristics while reducing the data volume compared to higher sampling rates.

Silence removal

Subsequently, the process eliminates non-essential sections of the audio file for classification or analysis. This involves setting an amplitude threshold; segments exceeding this threshold are identified as silent and removed. Removing silence reduces data size and enhances quality by eliminating extraneous elements from the signal.

Segmentation

Following silence removal, the audio signal is segmented into equal-sized units. Each 4-second segment of the audio file is subdivided into frames using a Hamming window with a 50 % overlap and each audio frame has a duration of one second as shown in Fig. 2. The use of a 4-second window is inspired by empirical evidence from previous studies that the isolated event duration of coughs usually falls between 0.5 and 3 s. This duration ensures the full capture of isolated bursts of cough and potentially the accompanying inspiratory phases or multiple consecutive bursts of coughs, with a moderate input size acceptable for downstream processing. It also effects a trade-off between resolution in the time domain and contextual information important in accurate acoustic feature extraction. This segmentation approach aligns with methods used in recent papers in the literature on COVID-19 detection from coughs and ensures comparability and resilience. Segmentation is overall a primary preprocess that allows modeling of the temporal dynamics and supports the extraction of discriminative localized features important in reliable audio classification.

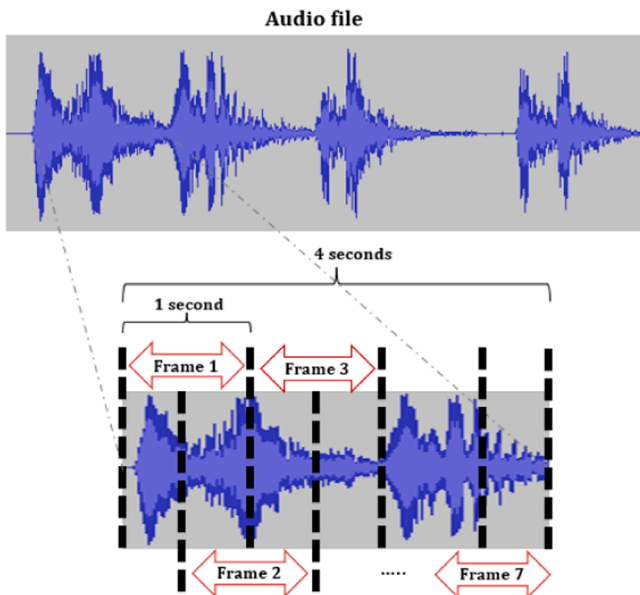


Fig. 2. Audio segmentation.

Background noise filtering

Audio signals often include interference from background noise, which can compromise the accuracy of classification models. Filtering techniques are applied to reduce or eliminate these auditory disturbances, such as the high-pass filter, resulting in a cleaner signal that retains only the most relevant information necessary for machine learning tasks.

Normalization

Following noise filtering, audio segments are normalized to ensure their feature values fall within a consistent range. This step is essential for maintaining feature homogeneity, enabling classification models to interpret data accurately. Additionally, normalization minimizes the impact of amplitude or intensity fluctuations that may occur across different portions of the audio signal. The amplitudes of the audio waveforms were adjusted using the formula described by Eq. (1)

$$A_{norm} = A/A_{max} \quad (1)$$

where, A represents the original amplitude of the audio. A_{max} represents the maximum amplitude. A_{norm} represents the normalized amplitude.

Features extraction

Various features can be extracted from an audio signal, but it is essential to focus on those most relevant to cough sound classification. We primarily use Mel Frequency Cepstral Coefficients (MFCC) [1,7,56] and Mel Filter Banks (MFB) methods [27,36], both of them are well-established methods in speech recognition and audio signal processing.

Mel frequency Cepstral coefficients

A common characteristic in the area of automatic sound identification is the MFCC [7]. Based on the Mel frequency scale, it represents the short-term power spectrum of a signal. The MFCC is motivated by the human hearing system.

The calculation of the MFCC parameters [16] uses a non-linear frequency scale.

The Mel frequency scale M [48] is defined by

$$M(f) = 2595 \log(1 + f/700) \quad (2)$$

where, f represents the frequency in Hz. $M(f)$ represents the frequency according to the Mel frequency scale.

The Fig. 3 shows the principle of calculation of the MFCC coefficients.

First, the Discrete Fourier Transform is applied on each segment and obtained with the formula [20]:

$$\sum_{t=0}^{N-1} y_i[t] w(t) e^{(-2\pi i k t / N)} \quad k = 0, \dots, N-1 \quad (3)$$

where, N is the number of samples in frame. $y_i[t]$ is the cough signal. $w(t)$ is the window function. $Y(k)$: DFT bin at frequency k

The signal spectrum is filtered by triangular bandpass filter on each cough signal's DFT output.

Mel filter bank

The Mel filter bank [36] is constructed by creating a series of triangular filters distributed in accordance with the Mel perceptual scale. Initially, the minimum and maximum frequencies of the signal spectrum are converted into the Mel scale. Subsequently, N_{mel} frequency points, uniformly distributed within the interval $[f_{min}, f_{max}]$, are generated and then converted back to the Hertz domain. Using these points, triangular filters illustrated in Fig. 4 are constructed, forming a matrix.

The filters are designed to weight the signal with values ranging from 0 to 1, maintaining the original signal where the weight is equal to 1. Their structure reflects the nonlinear distribution of bands on the Mel



Fig. 3. Calculation of MFCC.

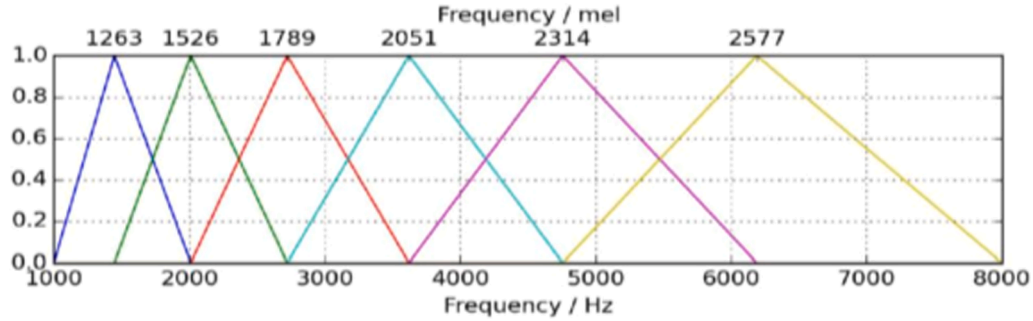


Fig. 4. Distributed filter bank according to the MEL scale.

scale compared to the Hertz domain, with narrower filters at lower frequencies and broader filters at higher frequencies. Each segment of the signal's spectrum is then multiplied by the appropriate triangular filter, and the logarithm of the resultant values is computed, finalizing the filter bank's creation. The resulting Mel filter bank is a matrix with rows corresponding to the employed frequency bands and columns corresponding to the segments produced by the Short-Term Fourier Transform obtained with the formula:

$$N_{\text{frequency}} = L/2 + 1 \quad (4)$$

where, L is the length of the FFT applied.

Data validation

After the feature extraction process, audio files are systematically organized into sub-folders based on their classification type. Each file comprises extracted features such as Mel Filter Bank (MFB) and Mel Frequency Cepstral Coefficients (MFCC), along with labels indicating either a genre or type of sound event. This results in well-structured and comprehensively labeled datasets that facilitate efficient training of classification models. Subsequently, a data validation phase is conducted to ensure the integrity of the datasets prior to model training. During this phase, all generated JSON files are examined for invalid samples, specifically checking for the presence of null values or zeros in the extracted features, which could compromise data quality. By identifying and excluding such anomalies, this step guarantees that only valid and reliable data are fed into machine learning algorithms, thereby preserving the robustness and accuracy of the models.

Data augmentation

The Synthetic Minority Over-sampling Technique, SMOTE, is a method used to address class imbalance issues within a dataset [45,49]. This approach enhances the representation of the minority class by artificially increasing its size, typically through the creation of synthetic instances between closely related samples. By applying this technique, SMOTE helps to mitigate bias during model training by balancing class distributions, ultimately leading to improved performance and enhanced generalization capabilities.

Data split

Dividing data into training and testing subsets is a crucial step in the evaluation of machine learning models. In this study, 80 % of the dataset is allocated for training, while the remaining 20 % is reserved for testing. The training subset enables the model to learn the underlying patterns and relationships within the data, while the test subset, which is only utilized at the final stage, evaluates the model's performance on

previously unseen data. This approach ensures that the model generalizes effectively to new data and provides a thorough and unbiased assessment of its performance.

Classification models

The classification of COVID-19-related cough sounds presents significant computational challenges due to the complex acoustic signatures involved. To address this, we conducted feature extraction on all audio samples prior to model input, using transformed acoustic representations rather than raw signals in our comprehensive comparative analysis of five machine learning architectures, each selected for their distinct theoretical advantages in audio pattern recognition:

- Convolutional Neural Networks (CNNs): These were prioritized based on their demonstrated efficacy in processing time-frequency representations, particularly their ability to extract hierarchical spatial features from spectrogram inputs through localized filter operations.
- Recurrent Neural Networks (RNNs): Implemented to capture temporal dynamics in cough sequences, as gated architectures have shown superior performance in modeling time-dependent bioacoustics patterns.
- Multilayer Perceptrons (MLPs): selected as a baseline for deep learning with fully connected architectures, offering flexibility in mapping non-linear relationships in handcrafted features.
- Random Forest (RF) classifiers: Incorporated as a robust, interpretable ensemble method for tabular features like MFCCs. Their resistance to overfitting and ease of implementation make them a strong traditional baseline.
- XGBoost ensemble methods were chosen as a state-of-the-art gradient boosting implementation, recognized for optimizing performance on structured acoustic features through sequential weak learner correction mechanisms

Convolutional neural network

CNN is a deep learning algorithm and works very effectively with image recognition and processing. Considering audio classification, CNNs can be applied to the spectrogram or other 2D representations of an audio signal [16,25]. Commonly, the architecture for CNN would include convolutional layers, pooling layers, and fully connected layers. The convolutional layers apply filters to the input data to extract the relevant features. The pooling layers reduce the spatial dimensions of the data, and the final classification is done by a fully connected layer. The key reasons behind gaining popularity are that CNNs can learn

hierarchical features automatically from raw data without much manual feature engineering [9,64]. In this work, the proposed CNN model has been designed to learn the spatial characteristic of cough sound. The complete layer-wise architecture, including parameter sizes and computational complexity (FLOPs), is summarized in Table 2. Architecture from input layer, followed by four convolutional layers with 128 and 256 numbers of filters accordingly; include after each MaxPooling, Batch Normalization, and ReLU activation. Resulting in feature maps that are flattened and passed through a fully connected Dense layer, a Dropout layer to prevent overfitting, and finally a SoftMax output layer for classification into three classes. This model was trained using the Adam optimizer, with a learning rate of 0.0004 and a categorical cross-entropy loss function, utilizing early stopping to prevent overfitting [39].

Random forest

Random Forest (RF) is an ensemble learning technique that is widely utilized in classification and regression tasks due to its robustness, predictive accuracy, and efficacy in handling high dimensional datasets [25,59]. It functions by constructing a large number of decision trees, each trained on a randomly selected subset of the training data and features. This randomization introduces diversity across trees, thereby enhancing the model's generalization ability and reducing the likelihood of overfitting. In the context of regression, the final prediction is obtained by averaging the outputs of all individual trees, which stabilizes the overall model prediction.

In this study, the RF model was configured using the following hyperparameters: a maximum tree depth of 8, a limit of 5 features considered per split, a minimum of 5 samples required to split an internal node, and a total of 500 decision trees comprising the ensemble. These settings were chosen to strike a balance between model complexity and generalization capability. The Mean Squared Error (MSE) was employed as the loss function to assess regression performance, as it quantifies the average squared difference between predicted and true target values. Furthermore, the ensemble averaging mechanism contributes to predictive stability by reducing variance, a defining strength of the Random Forest methodology.

Recurrent neural network

RNNs are neural network architectures exclusively functional over sequential data, thus they are most naturally fit for temporal or ordered-input tasks, such as time series prediction, speech recognition, and natural language processing. Owing to their recurrent nature, the network maintains an internal hidden state memory that keeps track of previous time steps, thereby enabling it to capture temporal dependencies across input sequences [29]. This characteristic renders

RNNs particularly well-suited for analysing the temporal patterns present in cough sound signals, where subtle acoustic variations may serve as indicators of health conditions such as COVID-19.

In this study, the model architecture is based on a purely RNN-based design incorporating a Long Short-Term Memory (LSTM) layer for effective sequence modelling. The input is first processed through an LSTM layer with 128 units, capable of capturing long-range temporal dependencies. Its output is passed to a fully connected Dense layer with 128 units, followed by a Dropout layer with a rate of 0.3, introduced to mitigate overfitting. The model concludes with a softmax output layer, which generates class probability distributions for multi-class classification tasks. Model training was performed using the sparse categorical cross-entropy loss function, appropriate for integer-encoded class labels, and optimized with the Adam optimizer at a learning rate of 0.000145. To avoid overfitting, early stopping was implemented based on validation loss monitoring.

These architectural and training configurations were empirically validated and fine-tuned to achieve a balanced trade-off between modelling capacity and generalization performance. The design prioritizes interpretability and reproducibility, aligning with established best practices in deep learning applications for sequential audio data.

Multi-layer perceptron

MLPs are feedforward neural networks composed exclusively of fully connected layers [45]. A typical MLP architecture includes an input layer, one or more hidden layers, and an output layer. Information flows unidirectionally from one layer to the next, with each neuron in a given layer connected to all neurons in the subsequent layer. A nonlinear activation function, commonly the Rectified Linear Unit (ReLU), is applied to the weighted sum of inputs at each neuron. The weights of these connections are optimized during training using the back-propagation algorithm, in conjunction with a gradient-based optimizer. MLPs are widely used in supervised learning tasks such as classification and regression and have demonstrated strong performance across various domains.

In the present work, the MLP architecture consists of three fully connected (Dense) hidden layers. Each hidden layer uses the ReLU activation function to introduce non-linearity and improve learning capacity. The final hidden layer contains 384 units, followed by a Dropout layer with a rate of 0.4 to reduce overfitting. The output layer uses a softmax activation function to generate class probabilities suitable for multi-class classification. The model is compiled with the categorical cross-entropy loss function. Training is conducted using the Adam optimizer with a learning rate of 0.0001. To further prevent overfitting, early stopping with a patience of 10 epochs is applied based on validation loss monitoring. These architectural and training choices were

Table 2
Layer wise architecture for CNN model.

Layer Name	Type	Output Shape	Kernel Size / Stride	Params	FLOPs
Input	-	(64, 64, 1)	-	0	0
Conv2D_1	Conv2D	(64, 64, 128)	(3 × 3) / (1 × 1)	1280	~67.1 M
MaxPool_1	MaxPooling2D	(32, 32, 128)	(2 × 2) / (2 × 2)	0	negligible
BatchNorm_1	BatchNorm	(32, 32, 128)	-	512	negligible
Conv2D_2	Conv2D	(32, 32, 128)	(3 × 3) / (1 × 1)	147,584	~33.6 M
MaxPool_2	MaxPooling2D	(16, 16, 128)	(2 × 2) / (2 × 2)	0	negligible
BatchNorm_2	BatchNorm	(16, 16, 128)	-	512	negligible
Conv2D_3	Conv2D	(16, 16, 128)	(3 × 3) / (1 × 1)	147,584	~16.8 M
MaxPool_3	MaxPooling2D	(8, 8, 128)	(2 × 2) / (2 × 2)	0	negligible
BatchNorm_3	BatchNorm	(8, 8, 128)	-	512	negligible
Conv2D_4	Conv2D	(8, 8, 256)	(3 × 3) / (1 × 1)	295,168	~33.5 M
MaxPool_4	MaxPooling2D	(4, 4, 256)	(2 × 2) / (2 × 2)	0	negligible
BatchNorm_4	BatchNorm	(4, 4, 256)	-	1024	negligible
Flatten	-	(4096,)	-	0	0
Dense_1	Dense	(256,)	-	1048,832	~1.05 M
Dropout	Dropout	(256,)	-	0	negligible
Dense_2	Dense	(3,)	-	771	~771
Total				1643,779	~152 M FLOPs

empirically selected to balance learning efficiency and generalization performance Table 3.

Extreme gradient boosting model

XGBoost is another very popular gradient boosting technique with a very wide application base in machine learning, including classification and regression tasks [13]. The XGBoost algorithm is a tree-based gradient boosting model that adds models stage-wise to create an ensemble [24]. It adds one model after another with the intent that every additional model tries to correct errors committed by previously added models. XGBoost introduces several key modifications over previous gradient boosting methods, such as handling missing values in input data, more regularized model formulation, and reducing problems with overfitting. Also, model training is accelerated through parallel processing techniques and a more efficient optimization approach.

In this study, the XGBoost classifier was optimized with the following hyperparameters: a maximum tree depth of 10 ($\text{max_depth} = 10$), a learning rate of 0.0475 ($\text{learning_rate} = 0.0475$), and 450 trees ($\text{n_estimators} = 450$). Additional regularization and generalization were introduced using $\text{subsample} = 0.8$, which controls the fraction of training instances used per tree, and $\text{colsample_bytree} = 1.0$, which sets the proportion of features considered when building each tree. The model objective was defined as multi: softprob for multi-class probability estimation, with the number of classes set to 3 ($\text{num_class} = 3$), corresponding to COVID, healthy, and non-cough labels. Model evaluation was guided by mlogloss and merror metrics, and early stopping was employed with a patience of 10 rounds to prevent overfitting.

This configuration was selected based on empirical validation performance and offers a robust balance between learning capacity and generalization. Bagad et al., [6].

Metrics

Testing data on the efficiency of the machine learning model is one of the major stages in its development and perfecting. It is based on a set of standardized metrics expressing the performance of the model, showing its merits and shortcomings [34].

Among the different metrics used in machine learning, accuracy, precision, recall, the F1 score, and confusion matrix are some of the most common metrics for classification. The current study used several of these metrics in the performance evaluation of the models developed.

Accuracy is a measure of the overall performance of a classifier. It is the number of proper predictions divided by total instances. Mathematically, it is expressed as:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FN + FP) \quad (5)$$

Where: TP (True Positive) represents the correctly identified positive cases. TN (True Negative) represents the correctly identified negative cases. FP (False Positive) represents the cases incorrectly identified as positive. FN (False Negative) represents the cases incorrectly identified as negative.

The precision of a classifier is defined as the ratio of correct positive predictions to the total number of positive predictions. It is calculated as follows [40]:

$$\text{Precision} = TP / (TP + FP) \quad (6)$$

Recall, also called sensitivity, measures the proportion of correctly

identified positive cases among all actual positive cases. Its formula is [40]:

$$\text{Recall} = TP / (TP + FN) \quad (7)$$

F1 score is a metric that combines precision and recall. It represents the harmonic mean of these two metrics, thus providing a balance between them. An optimal F1 score is 1, while the worst score is 0. It is calculated as follows [40]:

$$F1 - \text{score} = 2 * (\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision}) \quad (8)$$

In addition to these scalar metrics, the confusion matrix provides a tabular summary of model predictions, showing the distribution of predicted versus actual classes. It serves as the foundation for computing all of the above metrics.

These different metrics will enable thorough and detailed evaluation of model performance. This will help in finding out the exact strengths and weaknesses of the classifier, thus helping in the further development process by achieving more accurate and reliable results.

Model optimization

Hyperband tuner presentation

Hyperparameter optimization is a critical factor in determining the performance of machine learning models [13], particularly for complex architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and multilayer perceptron (MLPs). Key hyperparameters, including layer sizes, learning rates, and regularization factors, significantly influence a model's ability to generalize to unseen data. However, manually optimizing these parameters is often an expensive and time-intensive process due to the high-dimensional search spaces and the unpredictable interactions between hyperparameters. While traditional methods such as Grid Search are feasible for models with low computational demands, such as Random Forests, they become impractical for deep learning models due to the exponential growth in computational cost. In this context, the Hyperband Tuner, introduced by [32], emerges as an efficient alternative by intelligently allocating resources to evaluate hyperparameter configurations.

The Hyperband algorithm [13], based on the Successive Halving strategy, optimizes hyperparameter exploration and exploitation. It begins by defining a resource budget, such as training epochs or dataset sample sizes, and evaluating a large set of configurations with minimal resources. In subsequent iterations, resources are progressively allocated to the best performing configurations, eventually focusing on one or a few optimal combinations. This iterative approach allows Hyperband to explore an extensive search space while limiting computational costs for less promising configurations. Unlike Grid Search, which allocates equal resources to all configurations, Hyperband prioritizes promising combinations early in the process, making it particularly suitable for advanced machine learning models. Moreover, its flexibility lies in its independence from assumptions about the relationships between hyperparameters and model performance, ensuring its applicability across diverse scenarios. This innovative optimization technique thus strikes a balance between efficiency and accuracy, making it a valuable tool for tuning hyperparameters in complex models.

Optimization strategy for our models

The optimization of the models, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Multilayer perceptron (MLPs), and Random Forests (RFs), was conducted using techniques tailored to the specific characteristics of each algorithm. For neural network models (CNNs, RNNs, and MLPs), hyperparameter tuning was performed with Keras Tuner [50], utilizing the Hyperband algorithm to efficiently explore the search space by adaptively allocating computational resources [42]. The search space encompassed the number of filters for convolutional layers (ranging from 32 to 256), the number of units in dense layers (ranging from 64 to 512), dropout rates

Table 3
Confusion Matrix.

		Actual values	
		Positive	Negative
Predicted values	Positive	TP	FP
	Negative	FN	TN

(between 0.3 and 0.5), and learning rates (from 10^{-5} to 10^{-2} on a logarithmic scale). The optimization process was conducted over a maximum of 50 epochs, with Hyperband employing a reduction factor of 3. Early stopping was implemented to monitor either accuracy or loss on the validation set, with a patience threshold of 5 epochs to mitigate overfitting.

For Random Forest models, grid search, was used for hyperparameter optimization. The parameters considered included the number of trees (100, 200, 300), the maximum tree depth (None, 10, 20, 30), the minimum number of samples required to split a node (2, 5, 10), and the minimum number of samples required to form a leaf node (1, 2, 4).

Adding more trees increases model stability, but after a certain threshold, performance gains become marginal. Setting unlimited depth allows trees to fit freely, while restricting depth helps prevent overfitting. The minimum number of samples to split a node regulates complexity by avoiding unnecessary splits on small samples. Similarly, the minimum number of samples required to create a leaf node prevents overly specific leaves that could lead to overfitting. To reduce computational cost during the tuning phase, 3-fold cross-validation was used for Random Forest hyperparameter search only. However, all final performance evaluations were conducted using 5-fold cross-validation across all models and features (MFCC and MFB), as detailed in Appendix A.

All models were optimized with validation accuracy as the primary metric, reflecting a consistent approach to identifying optimal hyperparameter settings for both MFB and MFCC features in the context of cough sound classification. Additionally, variations in model architectures were explored within the defined search spaces, including the number of convolutional layers for CNNs and RNNs and the specific configurations of layers in MLPs.

Analysis of other optimization methods

In our experiments, Hyperband demonstrated superior performance compared to other hyperparameter optimization techniques, such as random search and Bayesian optimization, particularly for complex neural network models. For convolutional neural networks (CNNs), Hyperband [55] achieved an optimal solution in approximately 60 % of the time required by random search while delivering a marginally improved accuracy of 1.5 %. Although Bayesian optimization [33] performed comparably to Hyperband in terms of final accuracy, it demanded approximately 20 % more computational time.

For the Random Forest model, the differences between optimization methods were less pronounced, supporting the choice of grid search due to its simplicity and efficiency, especially given the lower computational requirements of this model type. These findings underscore the importance of selecting hyperparameter optimization strategies that align with the specific characteristics and computational demands of the model. Tailoring the approach to the model type not only enhances efficiency but also ensures optimal performance for machine learning tasks.

Ensemble learning approach

Ensemble learning was performed to enhance the performance of the models after observing an overfitting problem with the independent models during the initial CNN, MLP, RNN, Random Forest (RF), and XGBoost training, which was due to the small size of the database. The principle of this method is to combine multiple models with the intention of leveraging their complementary strengths and reducing their weaknesses to ensure better robustness and generalization of predictions.

Ensemble learning strategy

The methodology proposed in this study employs an ensemble learning strategy grounded in an aggregation framework that combines predictions from multiple models. Specifically, the stacking approach is

used, where a meta-model is trained to generate the final predictions by synthesizing outputs from base models. This approach reduces model-specific errors by learning to assign weights to predictions based on their relative importance.

The base models incorporated into this strategy include a Convolutional Neural Network (CNN), a Multi-Layer Perceptron (MLP), and a Recurrent Neural Network (RNN). These models were independently pre-trained on cough sound datasets, and their respective predictions on both training and test datasets were recorded. The predicted probabilities from the base models were then concatenated to train a Logistic Regression meta-model, which optimally combines these predictions to improve the system's overall performance. Unlike model averaging or fixed-weighting approaches, stacking enables the meta-model to adapt dynamically to the strengths and weaknesses of the individual models, thereby enhancing prediction accuracy and robustness.

Meta-model details

The fusion mechanism employed in this work is based on a stacking approach [6], wherein predictions from multiple base models are combined to form a more accurate and robust model. Specifically, three base models were utilized: a CNN, an MLP, and an RNN. These models were pre-trained and subsequently loaded from saved files.

Initially, a JSON file containing acoustic features represented as Mel Filter Banks and their corresponding labels is loaded. The dataset is then partitioned into training and test sets, with 20 % reserved for testing, and the class distribution is balanced.

Probabilistic predictions are then obtained from the CNN, MLP, and RNN models for both training and test sets. These predictions, derived from the penultimate layers of the models before the SoftMax activation (in multi-class classification), are concatenated along the feature axis. The concatenated outputs, referred to as meta-features, serve as inputs for training an additional model, the meta-model.

Logistic Regression [51] is selected as the meta-model, implemented using the Logistic Regression class from the scikit-learn library. This linear classification model accepts the concatenated predictions from the three base models as inputs and learns optimal weights to minimize classification errors. Default parameters are employed, including L2 regularization, which is one of the methods to combat overfitting by adding a penalty term to the model's loss function so that it penalizes large coefficients. This allows better generalization to new data. It also uses a linear optimization method and is effective for small datasets. Once trained, the meta-model leverages the information provided by the base models to generate the final predictions on the test data.

Model selection criteria for the ensemble

The ensemble incorporates the CNN, MLP, and RNN models because of their complementary strengths in feature extraction from cough signal data. CNNs are particularly adept at extracting local and spatial features through convolutional layers, making them highly effective for detecting repetitive patterns in Mel filter banks. In contrast, MLPs, as flexible and generic models, excel in capturing nonlinear relationships among diverse features, thereby complementing the feature extraction capabilities of CNNs. RNNs, particularly those employing LSTM cells, specialize in modeling temporal sequences, which is crucial for capturing the intrinsic temporal dynamics of acoustic signals such as coughs.

By leveraging the distinct strengths of these three models, the integrated approach provides the meta-model with rich and diverse feature representations, enabling it to combine these inputs optimally and achieve superior classification performance.

Results

Experimental results with original dataset

MFB results

The features of MFB demonstrated significant differences in the model's performance before any optimization of hyperparameters. Notably, the CNN achieved exceptional results, with an accuracy of 92.28 % and high metrics across the board: precision at 93.05 %, recall at 92.24 %, and F1 score at 92.63 %. This highlights the CNN's ability to extract meaningful patterns from the MFB features even without optimization.

The multilayer perceptron model (MLP) and the random forest (RF) also delivered respectable test accuracies of 89.43 % and 90.06 %, respectively. The RF model, in particular, showed strong consistency, with closely aligned results between training and testing, underlining its reliable generalization capabilities. The MLP also performed well, with its precision (89.53 %), recall (89.33 %), and F1 score (89.43 %) aligning closely with its accuracy.

The XGBoost model achieved balanced performance, with an accuracy of 86.66 %, precision of 87.20 %, recall of 86.70 %, and an F1 score of 87.00 %. These results indicate stable and consistent performance across metrics. Conversely, the recurrent neural network (RNN) exhibited the weakest performance, with an accuracy of 72.56 % and a significant drop in generalization. Its precision on the test set (87.00 %) sharply contrasts with its recall (73.00 %) and F1 score (78.00 %), suggesting potential overfitting during training.

Overall, these pre-optimization results emphasize that MFB features provide a solid foundation for the classification of cough sounds. Among the models tested, the CNN stands out as particularly promising, achieving the best balance of performance metrics even before hyperparameter fine-tuning [Table 4](#).

MFCC results

The table above presents the performance of various classification models using Mel Frequency Cepstral Coefficients (MFCC) before optimization. Among these models, the CNN stands out with an accuracy of 93.00 % on the test set. The model demonstrated a strong balance between precision (89.00 %) and recall (93.00 %), achieving a high F1 score of 91.00 %.

The multilayer perceptron (MLP) also performed well, with an accuracy of 90.53 % and excellent precision (96.84 %), paired with a recall of 88.58 %, resulting in an F1 score of 91.00 %. The random forest (RF) model showed slightly better recall at 92.27 % but slightly lower precision (88.50 %), yielding an F1 score of 89.97 %. Both the MLP and RF models provided strong and reliable performances, though slightly below that of the CNN.

The XGBoost model demonstrated balanced performance across metrics, with an accuracy of 87.51 %, precision and recall both at 87.40 %, and an F1 score of 87.40 %. In contrast, the recurrent neural network (RNN) displayed significantly weaker performance, with an accuracy of only 65.91 %, precision of 88.60 %, recall of 65.88 %, and an F1 score of 74.20 %. This underperformance suggests that the RNN may face challenges in effectively utilizing MFCC features, potentially due to overfitting or the unsuitability of this model for the task.

Overall, these pre-optimization results highlight the robust

performance of the CNN, MLP, and RF models when using MFCC features, with the CNN emerging as the most promising model [Table 5](#).

Comparison between MFB and MFCC

A comparative analysis of the results obtained using Mel Frequency Cepstral Coefficients (MFCC) and Mel Filter bank (MFB) features reveals notable differences in model performance prior to optimization. While MFCC features provided a reliable baseline, MFB features generally enhanced the overall performance of the models, particularly in terms of generalization and balance across metrics. The detailed performance of each model using both feature types is summarized in [Table 6](#).

The convolutional neural network (CNN) exhibited strong performance with both feature types. Using MFCC features, the CNN achieved an accuracy of 93.00 % and an F1 score of 91.00 %. However, with MFB features, the CNN showed slight improvements, reaching an accuracy of 92.28 % and an F1 score of 92.63 %. These results suggest that MFB features allowed the CNN to extract more discriminative information, thereby improving its classification ability.

Similarly, the multilayer perceptron demonstrated robust performance with both feature types. With MFCC features, the MLP achieved an F1 score of 91.00 %, while with MFB features, its F1 score was marginally lower at 89.43 %. Despite the slight difference, the MLP retained consistent accuracy across both feature sets, indicating that it effectively leveraged both types of features.

The random forest (RF) model showed consistent and reliable performance across the two feature types. With MFCC features, the RF achieved an accuracy of 92.14 % and an F1 score of 89.97 %, compared to an accuracy of 90.06 % and an F1 score of 87.00 % with MFB features. These results suggest that the RF model generalized slightly better with MFCC features, though the difference in performance is minimal.

The XGBoost model also demonstrated balanced performance across both feature sets. With MFCC features, the model achieved an accuracy of 87.51 % and an F1 score of 87.40 %. Similarly, with MFB features, its accuracy and F1 score were both 87.00 %, suggesting that XGBoost's performance was stable and not significantly impacted by the choice of feature type.

The most significant variation was observed with the recurrent neural network (RNN). Using MFCC features, the RNN exhibited the lowest performance, with an accuracy of 65.91 % and an F1 score of 74.20 %. However, with MFB features, the RNN's performance improved considerably, achieving an accuracy of 72.56 % and an F1 score of 78.00 %. This improvement highlights the potential of MFB features to mitigate some of the limitations observed with MFCC features for this model.

Experimental results with optimization

Before elaborating on the classification results, we draw attention to the optimization strategies that were implemented and that improved model performance. Hyperparameter adjustment was applied to all models using customized optimization techniques, as described in [Section 3.6](#). These optimizations improved generalization and robustness, yielding better and more reliable classification results.

MFB results

The experimental results demonstrate that the Mel Filter bank (MFB)

Table 4
Experimental results using MFB.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	92.28	93.05	92.24	92.63
MLP	89.43	89.53	89.33	89.43
RF	90.06	86.00	90.00	87.00
RNN	72.56	87.00	73.00	78.00
XGBoost	86.66	87.20	86.70	87.00

Table 5
Experimental results using MFCC.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	93.00	89.00	93.00	91.00
MLP	90.53	96.84	88.58	91.00
RF	92.14	88.50	92.27	89.97
RNN	65.91	88.60	65.88	74.20
XGBoost	87.51	87.40	87.40	87.40

Table 6

Performance comparison between MFB and MFCC features across different models.

Model	Features extracted	Accuracy (%)	F1-Score (%)
CNN	MFB	92.28	92.63
	MFCC	93.00	91.00
MLP	MFB	89.43	89.43
	MFCC	90.53	91.00
RF	MFB	90.06	87.00
	MFCC	92.14	89.97
RNN	MFB	72.56	78.00
	MFCC	65.91	74.20
XGBoost	MFB	86.66	87.00
	MFCC	87.51	87.40

features enabled strong performance across all models for the classification of cough sounds. Among the models evaluated, XGBoost achieved the highest performance, with an accuracy of 99.49 ± 0.0330 % on the test set and an exceptionally high F1 score of 99.55 %. These results highlight the robustness and precision of XGBoost when lever-aging MFB features.

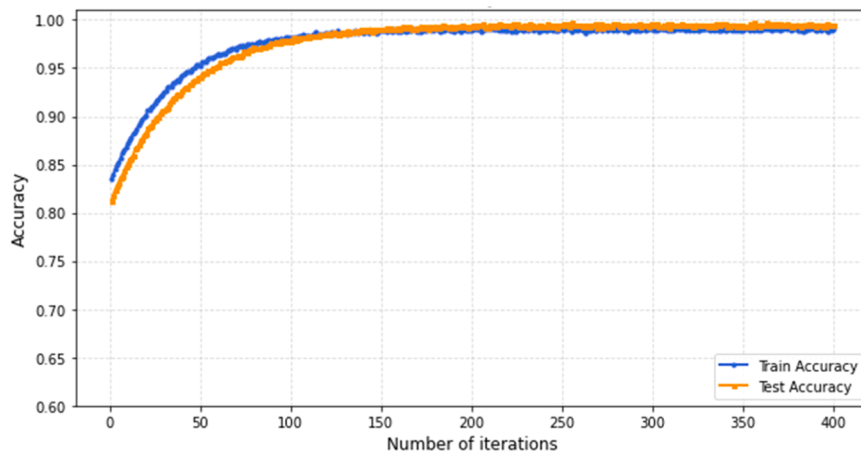
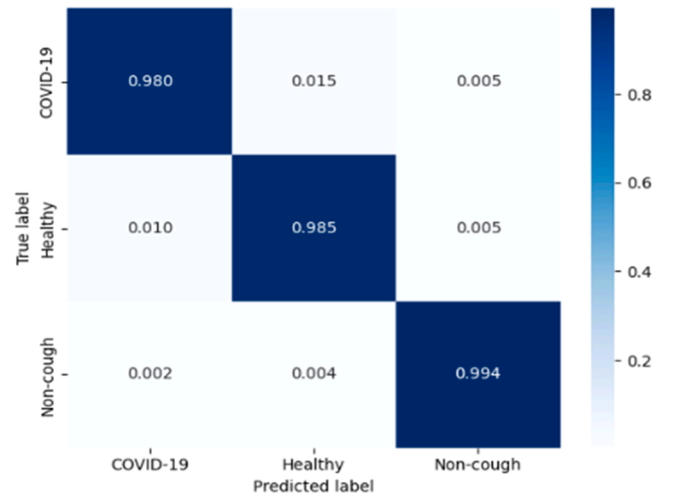
The evolution of accuracy during XGBoost training is illustrated in Fig. 5, which shows the convergence behavior of training and test accuracy. The rapid increase in accuracy during the first iterations, followed by stabilization at close to 99.5 %, under-lines the model's effectiveness in learning meaningful representations from MFB features.

The convolutional neural network (CNN) followed closely, with an accuracy of 98.41 ± 0.0018 % and an F1 score of 98.00 %. This demonstrates that the CNN also effectively extracted meaningful patterns from the MFB features, achieving results that are competitive with XGBoost.

The multilayer perceptron (MLP) further exhibited excellent performance, with an accuracy of 97.30 ± 0.0020 % and an F1 score of 97.00 %. These metrics underline the capability of the MLP to model complex relationships in the data, facilitated by the MFB features.

The random forest (RF) and recurrent neural network (RNN) models also delivered respectable results. The RF achieved an accuracy of 90.21 ± 0.0009 % and an F1 score of 68.00 %, while the RNN achieved an accuracy of 92.07 ± 0.0444 % and an F1 score of 90.00 %. These outcomes suggest that while the RF faced challenges in achieving high precision, the RNN capitalized on the MFB features to deliver balanced performance across metrics.

Fig. 6 presents the confusion matrix for XGBoost, clearly illustrating the model's ability to accurately distinguish between COVID-19 coughs, healthy coughs, and non-cough samples. The minimal number of mis-classifications observed further supports the strong performance metrics.

**Fig. 5.** Training and test accuracy of XGBoost model.**Fig. 6.** Confusion matrix of XGBoost model.

Overall, these findings underscore the effectiveness of MFB features in capturing the acoustic characteristics of cough sounds, enabling high quality discrimination across different model architectures. The consistently high performance across the majority of models highlights the potential of MFB features for cough sound classification tasks Table 7.

MFCC results

Mel Frequency Cepstral Coefficients (MFCCs) also demonstrated strong performance, albeit generally slightly less so than Mel Filter Banks (MFBs). Among the models evaluated, XGBoost once again stood out with an accuracy of 94.09 ± 0.0296 and an F1-score of 94.00 %, reaffirming its consistent performance in classification tasks based on these features.

The Random Forest (RF) model outperformed its MFB counterpart in terms of F1-score, achieving an impressive 94.61 %, despite having a

Table 7

Experimental results using MFB.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	98.41 ± 0.0018	99.43	99.42	98.00
MLP	97.30 ± 0.0020	99.36	99.36	97.00
RF	90.21 ± 0.0009	86.00	90.74	68.00
RNN	92.07 ± 0.0444	87.00	92.00	90.00
XGBoost	99.49 ± 0.0330	99.50	99.78	99.55

slightly lower accuracy of $90.24\% \pm 0.0513$.

The CNN and MLP models performed similarly, with respective accuracies of $92.59\% \pm 0.0020$ and $92.19\% \pm 0.0002$. The CNN achieved a slightly higher F1-score of 93.00% compared to the MLP's 92.00% .

Interestingly, the RNN model reported the highest accuracy, at $93.15\% \pm 0.0019$, but its F1-score was considerably lower at only 58.00% . This discrepancy likely indicates an issue with class imbalance or specific classification errors, as reflected in its relatively low recall (59.00%) and precision (58.00%).

Overall, MFCCs proved to be a robust foundation for classification, particularly for tree-based models such as XGBoost and Random Forest, which leveraged these features effectively to achieve a strong balance between precision and recall [Table 8](#).

Comparison between MFB and MFCC

When comparing the two feature extraction methods, Mel Filter Banks (MFB) demonstrated superior performance over Mel Frequency Cepstral Coefficients (MFCC) across most models and evaluation metrics. This performance gap was particularly evident in the case of deep learning models such as CNN and MLP. The detailed performance of each model using both feature types is summarized in [Table 9](#). The CNN model performed exceptionally well with MFB, achieving an accuracy of $98.41\% \pm 0.0018$ and an F1-score of 98.00% , compared to $92.59\% \pm 0.0020$ accuracy and a 93.00% F1-score with MFCC. Similarly, the MLP model significantly benefited from MFB, achieving an accuracy of $97.30\% \pm 0.0020$ and an F1-score of 97.00% , in contrast to $92.19\% \pm 0.0002$ accuracy and an F1-score of 92.00% with MFCC. These results show that MFB captures more relevant spectral information and better leveraged by deep models for the tasks.

In the case of tree-based models, such as RF and XGBoost, differences in performance for MFB and MFCC were less striking. For both feature types, XGBoost performed much better: it achieved an accuracy of $99.56\% \pm 0.0289$ and an F1-score of 99.55% with MFB, while these metrics were $94.09\% \pm 0.0296$ accuracy and an F1-score of 94.00% with MFCC. On the other hand, the RF showed dramatically worse results in terms of the F1-score in an MFB setting 68.00% in comparison to 94.61% in an MFCC setting, while the differences in accuracies were negligible: $90.21\% \pm 0.0009$ for MFBs and $90.24\% \pm 0.0513$ for MFCC.

These results show that both feature types are useful for tree-based models, though the choice may affect their abilities to handle the spectral information effectively.

The most surprising result was that MFB greatly improve the performance for the RNN model. While MFCC on the same model gave an F1-score of only 58.00% and an accuracy of $93.15\% \pm 0.0019$, the respective scores for MFB were 90.00% and $92.07\% \pm 0.0444$, respectively. That is a strong contrast, hence MFB capture better temporal dynamics relevant for sequential models like RNN.

In all, the best performance of many deep learning models was produced by the MFB, where the RNN's ability to preserve the critical spectral and temporal information was highly advantageous. However, MFCC showed respectable performance, especially in XGBoost and Random Forest tree-based models, which in turn under-lines that their usage is still relevant.

Table 8
Experimental results using MFCC.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	92.59 ± 0.0020	98.85	89.44	93.00
MLP	92.19 ± 0.0002	97.68	88.22	92.00
RF	90.24 ± 0.0513	95.31	94.55	94.61
RNN	93.15 ± 0.0019	58.00	59.00	58.00
XGBoost	94.09 ± 0.0296	95.00	94.00	94.00

Table 9

Performance comparison between MFB and MFCC features across different models with optimization.

Model	Features extracted	Accuracy (%)	F1-Score (%)
CNN	MFB	98.41	98.00
	MFCC	92.59	93.00
MLP	MFB	97.30	97.00
	MFCC	92.19	92.00
RF	MFB	90.21	68.00
	MFCC	90.24	94.61
RNN	MFB	92.07	90.00
	MFCC	93.15	58.00
XGBoost	MFB	99.49	99.55
	MFCC	94.09	94.00

Ensemble Learning results

The performance metrics of the MFB-based metamodel underscore its remarkable effectiveness in addressing the classification task, achieving an overall accuracy of 99.94% , a precision of 99.94% , a recall of 99.91% , and an F1-score of 99.92% . These outcomes highlight the metamodel's robust generalization capabilities, as it consistently delivering reliable predictions with minimal errors. The accuracy trend of the metamodel during training is depicted in [Fig. 7](#), demonstrating its rapid convergence towards near-perfect classification. Furthermore, the confusion matrix shown in [Fig. 8](#) provides a comprehensive view of the model's classification performance, confirming its ability to correctly identify the vast majority of samples with very few misclassifications. The consistently high accuracy across iterations further reinforces the model's stability and reliability in learning from MFB features, ensuring optimal performance on the dataset. The high recall and precision values indicate that the model is well balanced, excelling in identifying true positives while minimizing false positives, this balance ensures dependable performance across the dataset. Such exceptional results further validate the MFB feature extraction method as a powerful and scalable solution for enhancing classification tasks, particularly in the context of complex data challenges [Table 10](#).

Comparison with existing studies

Our works has been really outstanding, setting a new baseline in this important research area: COVID-19-classification cough sound using Mel-Frequency Bands. We developed and evaluated several machine learning models for the task using a diverse and large dataset that contained 13,860 audio samples from three different classes (COVID cough, healthy cough, and non-cough). The best result was achieved by XGBoost, which achieved an accuracy of 99.49% on the test set. This significantly outperforms the state-of-the-art results found in similar studies adopting various approaches, as summarized in [Table 11](#).

Our study is also distinguished by several key aspects:

- Diversity and size of the dataset: With 13,860 samples divided into three classes, our approach provides a strong basis for generalization, unlike some studies using smaller datasets.
- Variety of models: The exploration of multiple models (XGBoost, CNN, MLP, RF, RNN) demonstrates the robustness of MFB features across different machine learning approaches.
- Consistent performance: The high accuracy maintained between training and test sets for most of our models indicates excellent generalization, which is crucial for practical application.
- Class imbalance management: Despite significant class imbalance (684 COVID cough samples, 12,479 healthy cough samples, and 697 non-cough samples), our models exhibited robust performance, highlighting their effectiveness in real-world scenarios.

For example, [\[38\]](#) employed XGBoost with MFCC features and

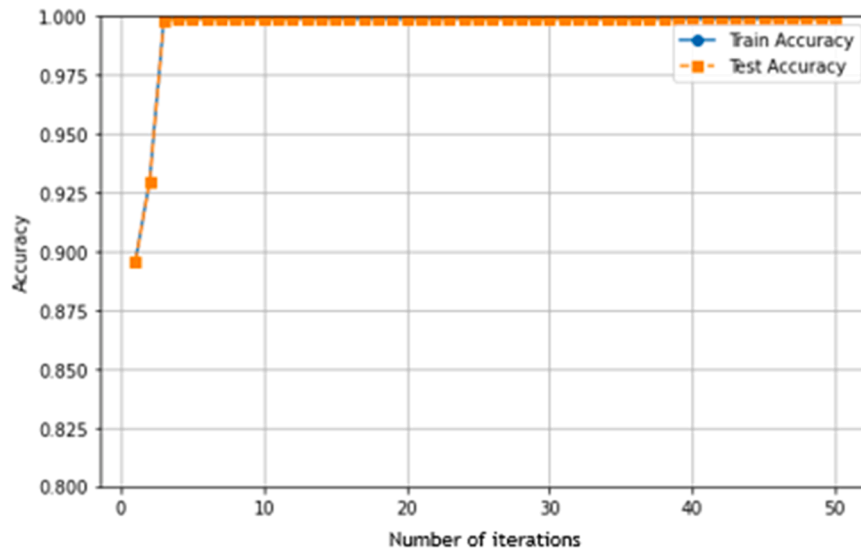


Fig. 7. Training and test accuracy of metamodel.

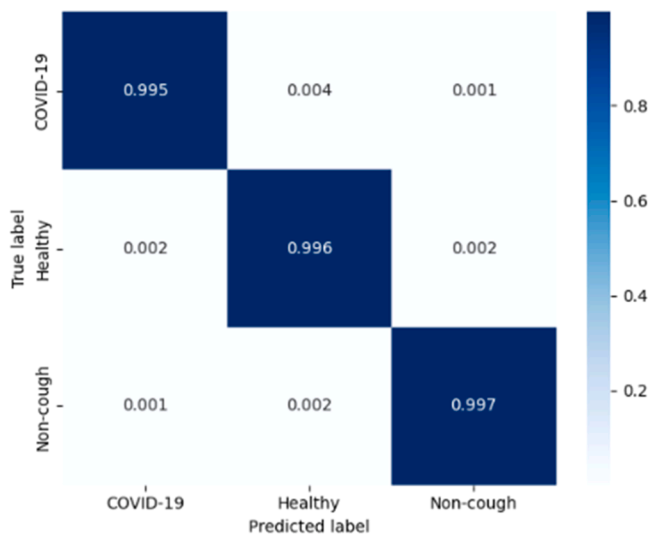


Fig. 8. Confusion matrix of metamodel.

Table 10

Ensemble learning results using MFB.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Meta-model	99.94	99.94	99.91	99.92

re-reported an accuracy of 97 %, which is slightly lower than the 99.49 % we achieved. Similarly, [45] utilized deep neural networks with MFCC features and achieved 97.5 % accuracy. [11] demonstrated a strong performance of 93.91 % using ResNet and MobileNet models with MFB features. In comparison, our XGBoost model surpasses their results significantly. [55] combined CNNs and SVMs in an ensemble model and achieved 97.8 %, while our results still exceeded their performance by a considerable margin.

Recent studies further underscore this trend. [26] combined the COUGHVID, Coswara, Cambridge datasets and used ResNet and DNN models with MFCC, Δ -MFCC, and spectral contrast features, achieving a sensitivity of 95 % and specificity of 96 %. Meanwhile, [58] employed YAMNet with pretrained vision transformers using Fractal Dimensions Coefficients, reporting a remarkable accuracy of 98.45 %. [29] applied

CNN architectures to MFCC features and reached an accuracy of 92.1 %. In another noteworthy case, [4] achieved a sensitivity of 89 % using time-frequency features with a deep learning framework.

When it comes to ensemble learning approaches, such as the one used in our work, combining CNN, MLP, and RNN models with MFB features, our results are unprecedented. Our model achieved a remarkable overall accuracy of 99.94 % on a balanced subset of the data. This performance demonstrates the robustness and generalization of our approach even under conditions of severe class imbalance. In comparison, studies such as [46], who employed ensembles of CNNs and RNNs, reported a maximum accuracy of 98.5 %, while Coppock et al. [3], using an ensemble of decision trees, achieved an AUC of only 0.88.

Furthermore, studies employing alternative features such as cepstrum-based features or other time-frequency representations, like those of [23] and [59], fall significantly short of our results. These studies reported accuracies of 93 % and 83.67 %, respectively. This comparison underscores the effectiveness of Mel-Frequency Bands (MFB) as the feature set in conjunction with advanced machine learning and ensemble models.

In conclusion, our work not only surpasses the results of similar studies in terms of accuracy, F1-score, and generalization but also highlights the potential of MFB features in COVID-19 detection. The strong validation of our ensemble learning approach establishes a new milestone for acoustic COVID-19 detection and opens promising perspectives for public health applications.

Discussion

Our research in COVID-19 acoustic detection has demonstrated significant advancements, particularly through the adoption of our XGBoost approach and ensemble learning method, both utilizing MFB features. These results are particularly remarkable given the 1:18 ratio between COVID and healthy samples, underscoring the robustness and real-world applicability of our models in class-imbalanced scenarios. The XGBoost model delivered an impressive accuracy of 99.49 % on the test set, while our CNN, MLP, and RNN-based ensemble learning model achieved an accuracy of 99.94 %. This is particularly remarkable given the almost 1:18 ratio between COVID and healthy samples, showcasing the robustness of our methods and their applicability to real-world scenarios, where positive cases are often a minority.

The superior performance of XGBoost can be attributed to its gradient boosting framework, which excels at handling structured tabular features such as MFB representations and is inherently robust to

Table 11

Summary of previous COVID 19 cough classification work.

Authors	Methods (classifier / features)	Results (%)	Advantages	Limitations
[37]	CNN / MFCC	Acc: 77.00	-Data augmentation through segmentation -Multi-image representation -Ensemble modeling	-Class imbalance -Variable audio quality -No clinical validation
[15]	4 dense neural network and 3 CNN / MFCC	Acc: 88.75	- Uses both cough and breathing sounds. - CNN-based model achieves good performance on public dataset. - Lightweight model suitable for mobile.	- No clinical validation or large-scale testing. - Limited diversity in training data.
[23]	Deep CNN Multiclass classifier / MFCC	Acc: 92.85	- Deployed as a mobile app. - Risk-averse AI design with multi-pronged mediators. - Handles cough vs. non-COVID differentiation.	- Not a clinical diagnostic tool. - Needs larger-scale labeled datasets for better generalization.
[11]	ResNet 6, MobileNet6, VGG7 and CNN4 / MFB	Acc: 93.91	- Uses large-scale datasets (FluSense, COUGHVID). - Applies transfer learning to improve generalization. - Outperforms baselines on DiCOVA and ComParE datasets.	- Mismatch between source and target datasets. - Does not use multi-modal features (e.g., breath, voice).
[35]	SVM / Power spectral density based on short-time Fourier transform and MFCC	Acc: 94.21	- Simple, interpretable signal processing approach.	- Small dataset - No external or benchmark validation.
[29]	DCNN / DAE, GFCC and IMFCC	Acc: 95.45	- Uses cough, breath, and speech signals for richer features. - Transfer learning and bottleneck features improve performance.	- Pretraining done on non-COVID-labeled datasets. - Real-world deployment not validated.
[28]	CNN, Poisson biomarker and 3 pre-trained ResNet50 / MFCC	Acc: 97.10	- Massive dataset. -Low-cost, scalable solution.	- Results need clinical validation. - Model trained on a specific dataset/timeframe.
Lella et al., [31]	1D CNN / Data De-noising Auto Encoder	Acc: 90.00	- Combines cough, breath, and voice. - Effective augmentation and 1D CNN model.	- Dataset relatively small. - Generalizability not tested across populations.
[2]	2D CNN / EMD and MFCCs	AUC: 97	- Clinically validated dataset. - Detects severity levels. - Adaptable for mobile/edge devices.	- Limited details on false positives/negatives. - Implementation complexity for field use.
[18]	SVM, KVM, RF / Common Short Term Features and MFCCs	AUC: 90	- End-to-end system from audio to diagnosis.	- Limited by dataset source compatibility. - No sensitivity/specificity breakdown.
[21]	LSTM (RNN) / Spec. Centroid, Spec. roll-off, ZCS, MFCC (+ $\Delta\Delta$)	Acc: 97.00	- Utilizes LSTM for temporal modeling. - RNN captures cough sound dynamics.	- No extensive validation dataset. - Modest performance metrics reported.
[38]	XGBoost / MFCC	Acc: 97.00	-Uses symbolic dynamics and MFCCs.	- Focused on synthetic and sustained vowel data.
[45]	DNN / MFCC	Acc: 97.5	- Innovative data undersampling. - Data from six continents. - Smartphone-compatible.	- May lack generalizability. - Imbalanced dataset. - No reported clinical trials.
[63]	Contrastive Learning / Feature Encoder and Random Masking	Acc: 83.74	- Uses ensemble learning with self-supervised features. - Competitive results on benchmark datasets.	- Limited transparency of pretraining data. - Ensemble complexity increases computation.
[59]	Most popular supervised models / Time-frequency features	Best Acc: Random Forest: 83.67	- Uses interpretable ML with cough segmentation. - Designed for real-time applications.	- Small dataset used - External validation lacking.
[4]	Deep learning model / Time-frequency features	Sensitivity: 89,00	- Focused on cough segmentation for enhanced classification. - Preprocessing improves signal clarity.	- Not peer-reviewed. - No reported test accuracy or metrics.
[30]	CNN / MFCC	Acc: 92.1	- Combines MFCC, spectrogram, and statistical features.	- Dataset limited to lab conditions. - Lacks real-world deployment testing.
[26]	Resnet and DNN / MFCC, $\Delta 2$ -MFCC, Δ -MFCC, and spectral contrast	Sensitivity: 95,00 Specificity: 96,00	- Combines multiple acoustic features. - Uses statistical modeling and neural nets. - Open access dataset.	- Performance may vary by mic type. - Not tested cross-culturally.

(continued on next page)

Table 11 (continued)

Authors	Methods (classifier / features)	Results (%)	Advantages	Limitations
[58]	YAMNet and pretrained vision transformer / Fractal Dimensions Coefficients	Acc: 98.45	- Uses Vision Transformer (ViT) and Grad-CAM for explainability. - Fractal features add interpretability.	- High computational cost. - Complex pipeline for clinical integration.
Hussain et al., [22]	Multi-layer ensemble deep learning (MLEDL) / framework (MFCC-MLP, Spectrogram-CNN, Chromagram-MLP)	Acc: 98	- High Accuracy - Non-Invasive - Robust Feature Selection	- Data Dependency - Noise Sensitivity - Computational Cost
Topuz, Kaya [60]	Super Learner (SL)-based ensemble model / Chroma vector, MFCC, Mel-frequency, Spectral, Poly-features, Tonnetz, Zero Crossing Rate (ZCR)	Acc: 74.4	- Feature diversity - Clinical applicability - Handles imbalanced data	- Computational complexity - Sensitive to noise
Ayappan, Anila [5]	Enhanced Deep Neural Network (EDNN) / Zernike Moments, Gray Level Co-occurrence Matrix,	Acc: 98.5	Error Minimization Non-Invasive High Accuracy	- Noise Sensitivity - Computational Cost - Overfitting Risk
Proposed work	XGBoost / MFB	Acc: 99.49	-Managing imbalance -Rigorous feature validation -Complete and fair evaluation -Flexibility and Extensibility	-Small dataset used -Increased training costs
Proposed work	Ensemble learning (CNN, MLP, RNN) / MFB	Acc: 99.94	-Managing imbalance Robust Ensemble	-Small dataset used - Increased complexity

noise and overfitting due to its use of regularized decision trees.

Meanwhile, the ensemble learning strategy leverages architectural complementarity: CNNs are effective at extracting spatial features from spectrogram-like data, MLPs capture abstract nonlinear relationships, and RNNs contribute temporal modeling capabilities. By combining these strengths, the meta-model generalizes better and offers enhanced resilience to variations in audio signals and patient specific factors.

This consistency, reinforced by rigorous cross-validation, confirms the generalization capability of our models while avoiding overfitting. The diversity of evaluated models also highlights the versatility of MFB features across different machine learning contexts.

Regarding the deployment potential in real-world telehealth applications, our proposed system presents several promising characteristics that align with practical healthcare requirements. The computational efficiency of our XGBoost model, combined with its exceptional performance, makes it particularly suitable for deployment in resource-constrained environments or mobile applications where low latency is crucial. The system's ability to process audio samples rapidly could enable real-time screening in telehealth platforms, allowing patients to receive immediate feedback during virtual consultations. From a privacy perspective, the acoustic analysis can be performed locally on user devices, minimizing the need to transmit sensitive health data to external servers. Additionally, the non-invasive nature of cough analysis makes it highly compatible with user-friendly interfaces, requiring only standard microphone capabilities available on most smartphones and computers. However, practical deployment would necessitate careful consideration of audio quality standardization across different devices, implementation of robust noise reduction algorithms, and development of clear user guidelines to ensure consistent recording conditions. Integration with existing electronic health records and compliance with healthcare data protection regulations would also be essential for widespread adoption in clinical telehealth environments.

While the large size of our dataset is a strength that allows for a comprehensive evaluation of our models, some limitations warrant further exploration. For instance, while class imbalance is well-handled in our study, investigating the performance of our models under even more extreme imbalance scenarios or across varying epidemiological contexts would add further depth. Similarly, understanding the impact of different recording conditions, equipment, and the emergence of new virus variants is critical for optimizing our model's real-world performance.

These challenges and future research opportunities present exciting possibilities, such as exploring hybrid architectures combining XGBoost and deep learning methods or employing advanced data augmentation techniques for the minority class. Clinical validation at a large scale,

with realistic positive-to-negative case ratios, is essential to confirm the practical effectiveness of these systems.

Beyond COVID-19 detection, the proposed methodology exhibits strong potential for generalization to other biomedical acoustic classification tasks. For example, respiratory illnesses such as asthma, pneumonia, or chronic obstructive pulmonary disease often present distinct audio signatures in cough or breathing sounds that can be modeled using the same MFB/MFCC-based preprocessing and ensemble learning framework. The modular design of the pipeline spanning noise filtering, feature extraction, and robust model stacking, allows seamless adaptation to different domains, provided domain-specific labeled audio data is available. This versatility opens the door for real-time health screening and other edge-AI applications.

Conclusion

This work provides a new benchmark in the area of COVID-19 acoustic detection, as evident from its outstanding performance against the high challenges this highly im-balanced dataset presents. By embedding the feature representations of the MFB into higher-order machine learning and ensemble learning techniques, our models achieve outstanding performances: 99.49 % by XGBoost and an ensemble learning model based on deep fusion of CNN, MLP, and RNN architectures achieved an out-standing accuracy of 99.94 %. These results show the robustness and applicability of our approach to real-world problems even in the presence of heavy class imbalance.

The consistent performance across training and test sets underlines the strong generalization capability of our models, while rigorous cross-validation further reinforces their reliability. This underlines the potential of MFB features as a versatile representation in diverse machine learning contexts.

Whereas class imbalance was an important challenge, our methodologies have considerably smoothed over the benchmark for the acoustic COVID-19 detection task.

To further advance this work, several promising directions are envisioned. First, we will investigate hybrid feature fusion strategies, such as attention-based mechanisms or weighted concatenation of MFCC and MFB features, to explore potential synergies while optimizing computational efficiency. Second, alternative class imbalance mitigation techniques, including adversarial generative networks and dynamic cost-sensitive learning, will be evaluated to enhance generalization on minority classes. Third, the ensemble framework could be refined through interpretable model stacking or automated model selection to balance performance and complexity. Additionally, we plan to incorporate self-supervised feature learning paradigms. These extensions aim

to address current limitations while maintaining the methodological rigor demonstrated in this study.

Future work might consider exploring hybrid architectures, such as using XGBoost to select features, combining it with deep learning models; investigate variability in re-cording conditions and how different environmental factors affect the performance of the systems; or using more informed data augmentation techniques in order to better emphasize minority classes. The final step will involve large-scale clinical validation under realistic case distributions to confirm their scalability and practicality. Amenable to these challenges, the proposed methods can then play a pivotal role in the management of current and future pandemics by facilitating scalable, accessible, and effective public health surveillance and triage.

CRedit authorship contribution statement

Abdellah Touhafi: Writing – review & editing, Validation, Supervision, Project administration. **Mohammed Jorio:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **Malika Alami Marktani:** Writing – review & editing, Validation, Supervision, Methodology. **Hiba Chougrad:** Writing – review & editing, Validation, Methodology, Conceptualization. **Btissam Bouzammour:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ghita Zaz:** Writing – review & editing, Validation, Supervision, Project administration, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jer.2025.08.017](https://doi.org/10.1016/j.jer.2025.08.017).

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